

NATIONAL HIGHER SCHOOL OF ARTIFICIAL INTELLIGENCE (ENSIA)

Algiers, Algeria

bacAidz

An End-to-End Machine Learning Platform for the Algerian Baccalaureate Journey

Personal Project Report

Pre-BAC, BAC Year, and Post-BAC: Data Science, Explainable AI,
and Full-Stack Engineering Applied to Algerian Secondary Education

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Preface

This project was born during the spring vacation of 2026, while I was a second-year preparatory (CP2) student at the National Higher School of Artificial Intelligence (ENSIA) in Algiers. During that break, I decided to deepen my understanding of machine learning by independently following a course on GeeksforGeeks — the kind of place where students with a genuine passion for the field enjoy spending their free time.

After two years at ENSIA, I had already built solid foundations in mathematics, including linear algebra, probability, statistics, and calculus, alongside strong programming skills in Python, Java, and C++. I had also gained experience in data manipulation and in building modern full-stack web applications using HTML, CSS, JavaScript, Figma, and the MERN stack through my web development coursework. At that point, I finally felt ready to combine everything I had learned into a single, meaningful project.

Like most beginners, my first instinct after completing the course was to search online for interesting machine learning projects. Unsurprisingly, the suggestions revolved around the usual beginner exercises: house price prediction, stock price prediction, Iris classification, and similar tutorials. While excellent for learning the fundamentals, these are also the projects that almost everyone builds by following a video. I wanted something more ambitious — something capable of solving a real problem that mattered to me personally.

Around that time, I noticed that many ENSIA students dedicated their Instagram accounts to producing educational content for Baccalaureate students. Watching them share advice, study tips, and motivation made me think about how I could contribute using the skills I had just acquired. That is when the idea came to me: *why not build an intelligent platform capable of predicting and analyzing the Algerian Baccalaureate?*

In Algeria, the Baccalaureate is far more than a national examination; it is a decisive milestone that shapes a student's entire academic and professional future, and it is surrounded by countless assumptions and stereotypes. Do Brevet results really influence Baccalaureate performance? Are trimester grades reliable indicators of final results? Does the chosen academic stream (*filière*) significantly affect performance? Do students within the same stream naturally form distinct profiles? Many students also struggle throughout the BAC year to evaluate their level, identify their strengths and weaknesses, and make informed decisions about their future orientation.

These questions convinced me that this was the right project to pursue. It addressed a real problem faced by hundreds of thousands of Algerian students every year, and it gave me the opportunity to apply machine learning to a local context while collecting and understanding

the data myself, rather than relying on Kaggle or other international datasets. Since this was my first complete data science project, I wanted to experience the entire pipeline — from data collection and generation, to preprocessing, analysis, model training, and deployment.

I spent nearly eight weeks working on this project. My productivity was not always consistent, as I was simultaneously completing my academic semester and preparing a visa application. Despite these challenges, I remained committed to bringing the project to completion.

From the very beginning, my goal was to transform the tools that machine learning offers into something that any Baccalaureate student, with no knowledge of computer science, could easily use and understand. This project is about discovering meaningful patterns hidden within Algerian Baccalaureate data, teaching those patterns to machines, and transforming them into an interactive application that students can explore and genuinely benefit from. This philosophy extended to every interface component, color palette, and page layout, all of which were designed with a specific audience in mind: a 17-year-old student living through the Baccalaureate journey.

I do not claim this platform to be perfect. As my first complete end-to-end machine learning application, there are undoubtedly areas that could be improved, along with limitations and biases that future work may address. Nevertheless, I firmly believe that the platform provides Baccalaureate students with valuable insights into their academic journey and helps them make more informed decisions through a unique and innovative approach.

The application follows a strategic structure, dividing the Baccalaureate journey into three major decision-making phases experienced by every Algerian student: **Pre-BAC**, **BAC Year**, and **Post-BAC**. Rather than treating the Baccalaureate as a single event, I chose to model it as a complete journey, where each phase presents its own challenges, questions, and decisions. The motivation behind each of these phases, inspired by my own experience as a Baccalaureate student, is detailed throughout this report.

Abstract

The Algerian Baccalaureate (BAC) is one of the most decisive milestones in the academic life of Algerian students, acting as the primary gateway to higher education. Despite its importance, students often lack data-driven tools to understand their own academic trajectory, anticipate their performance, or choose an appropriate higher-education orientation. Most existing educational technology either targets international markets with generic content or addresses isolated tasks such as grade prediction, without modeling the complete journey experienced by Algerian students.

This report presents **bacAidz**, an end-to-end machine learning platform designed specifically for Algerian Baccalaureate students. Rather than treating the Baccalaureate as a single prediction problem, the platform is structured around three successive phases of the student journey: **Pre-BAC**, which investigates whether previous academic results, study habits, and regional or cultural factors influence final BAC performance, using statistical analysis and K-Means clustering; **BAC Year**, the core of the platform, which predicts a student’s expected grade in each of the fifteen Baccalaureate subjects using subject-specific Random Forest models, explains every prediction with SHAP (SHapley Additive exPlanations), and provides simulation tools, personalized resource recommendations, automatically generated to-do lists, and an optimization-based study timetable generator; and **Post-BAC**, which recommends suitable higher-education domains and specific university programs using a hybrid system combining stream-specific Random Forest classifiers with a rule-based eligibility engine built on official admission criteria.

The project required the construction of original, representative datasets describing the Algerian educational system, since no public dataset offered the required scope. Data was collected through structured surveys and official educational resources (Ency-Education, DZ-Exams, Ministry of Education publications), then carefully extended through a controlled, validated generation process to reach approximately 1,500 to 2,000 student records per module. The resulting models, trained primarily using Random Forest algorithms, achieved an average coefficient of determination (R^2) of approximately 0.895 across the fifteen BAC Year subject models, and stream-specific Post-BAC orientation classifiers reaching 85–95% accuracy.

Beyond the machine learning pipeline, the project required designing and implementing a complete MERN-stack web application (MongoDB, Express.js, React.js, Node.js), including a RESTful API, JWT-based authentication, a MongoDB schema connecting users to their predictions across all three phases, and a culturally adapted, Arabic-first user interface tailored to a teenage Algerian audience.

This report documents the full engineering process behind bacAidz: from problem framing and data collection to exploratory analysis, model development, explainability, rule-based reasoning, backend architecture, and the resulting product. It also critically discusses the project's limitations, particularly regarding the use of synthetic data, and outlines a roadmap of future improvements.

Keywords: Machine Learning, Random Forest, Explainable AI, SHAP, Education Technology, Algerian Baccalaureate, MERN Stack, Clustering, Recommendation Systems, Genetic Algorithms.

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Chapter 1

Introduction

The Algerian Baccalaureate (BAC) represents one of the most significant milestones in the academic journey of Algerian students. Beyond being a national examination, it serves as the primary gateway to higher education and largely determines the academic and professional opportunities available to students after secondary school. Every year, hundreds of thousands of candidates prepare for this examination while facing considerable academic pressure, societal expectations, and uncertainty regarding their future.

Despite the abundance of educational resources available today, students often struggle to answer fundamental questions throughout their Baccalaureate journey. Before entering the final year of high school, many wonder whether their previous academic performance is truly representative of their future potential. During the Baccalaureate year, students frequently seek to identify their strengths and weaknesses, estimate their expected performance, prioritize their revision efforts, and manage their time effectively. Finally, once the examination is over, they face another critical challenge: choosing the higher education pathway that best matches their academic profile, interests, and predicted results.

While machine learning has demonstrated remarkable capabilities in discovering hidden patterns within data and supporting decision-making across numerous domains, its application to the Algerian educational context remains relatively limited. Existing educational applications often focus on isolated tasks such as grade prediction or simple orientation tools, without considering the complete academic journey experienced by students. Moreover, many available machine learning projects rely on publicly available international datasets, making them less representative of the realities and specific characteristics of the Algerian educational system.

1.1 Project Motivation

This project aims to address these limitations by proposing **bacAidz**, an end-to-end machine learning platform specifically designed for Algerian Baccalaureate students. Rather than treating the Baccalaureate as a single prediction problem, the platform models it as a complete journey composed of three successive phases: **Pre-BAC**, **BAC Year**, and **Post-BAC**. Each phase addresses a different set of challenges through the integration of data analysis, machine learning, explainable artificial intelligence, rule-based systems, and modern web technologies.

1.2 Objectives

The objectives of this project can be summarized as follows.

- Collect, generate, and organize representative datasets related to the Algerian Baccalaureate.
- Explore and analyze these datasets to identify meaningful academic patterns and relationships.
- Develop machine learning models capable of assisting students in predicting academic performance and supporting educational decision-making.
- Integrate explainability techniques to improve the transparency and interpretability of model predictions.
- Design and implement an intuitive web platform that transforms complex machine learning models into accessible tools for students with no prior knowledge of artificial intelligence.
- Provide additional intelligent services, including personalized recommendations, study planning tools, and higher education orientation assistance.

1.3 Contributions

The main contributions of this project extend beyond the development of predictive models. They include:

- the construction of original datasets tailored to the Algerian educational system, built from surveys and official examination resources rather than reused from public repositories;
- the design of an end-to-end intelligent platform combining supervised learning, unsupervised learning, explainable AI, and rule-based reasoning across three distinct decision-making phases;
- the development of a user-centered, Arabic-first interface specifically adapted to the needs, language, and culture of Algerian Baccalaureate students;
- a fully functional MERN-stack web application connecting all of the above into a coherent, deployable product.

1.4 Report Organization

The remainder of this report is organized as follows. Chapter 2 presents the overall architecture of the system, including the project organization, technologies, database design, and user interface philosophy. Chapters 3, 4, and 5 describe the three main phases of the application — Pre-BAC, BAC Year, and Post-BAC, respectively — detailing the datasets, exploratory

analyses, machine learning models, evaluation procedures, and user interfaces developed for each module. Chapter 6 discusses the backend architecture and software engineering aspects of the application. Chapter 7 discusses the limitations of the current system. Chapter 8 presents potential future improvements, and the report concludes with a final summary of the work.

Chapter 2

Project Analysis and Global Design

The development of this project followed a complete end-to-end engineering workflow, beginning with data collection and ending with the deployment of a fully functional web application. Before discussing each module in detail, this chapter provides a global overview of the different stages involved in the project's development.

The first stage consisted of identifying the most relevant features and collecting representative data describing the academic journey of Algerian Baccalaureate students. To achieve this, several data collection forms were designed and distributed among students. Additional information was gathered from reliable educational resources and official publications, including Ency-Education, DZ-Exams, official social media pages, national examination statistics, and official journals published by the Algerian educational authorities.

Although these sources provided valuable and generally reliable information, the collected data inevitably contained missing fields, inconsistencies, noise, and occasional inaccuracies. Furthermore, some information required by the machine learning models was simply unavailable. For this reason, a data generation phase was introduced to reconstruct the missing information as realistically as possible while preserving the characteristics of the Algerian educational system. The objective was not to artificially inflate the datasets, but rather to build representative datasets that accurately reflect the reality experienced by Algerian students.

Once the data collection and generation processes were completed, all datasets were converted into structured CSV files. A comprehensive preprocessing stage was then carried out, including data cleaning, handling missing values, removing duplicate records, detecting and treating outliers, reducing noise, and validating the consistency of the collected information. The final datasets contained approximately 1,500 to 2,000 representative student records, depending on the module.

The project was then organized into three independent modules corresponding to the three major phases of the Algerian Baccalaureate journey: Pre-BAC, BAC Year, and Post-BAC. For each module, dedicated Jupyter notebooks were developed to conduct Exploratory Data Analysis (EDA), investigate meaningful patterns within the data, answer the educational questions identified at the beginning of the project, engineer relevant features, compare different machine learning algorithms, train and evaluate the selected models, and validate their predictions using multiple representative student profiles.

After completing the machine learning pipeline, the focus shifted toward software devel-

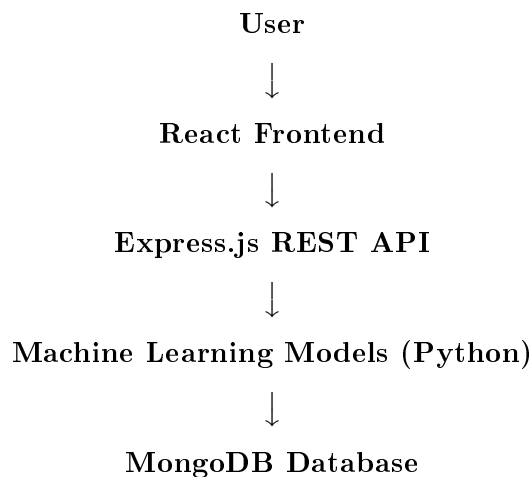
opment. The objective was to transform the trained models into an interactive platform that could be easily used by students with no background in artificial intelligence. The application was therefore developed using the MERN stack, providing a modern, responsive, and scalable architecture.

The graphical user interface was first designed using Figma to establish a consistent visual identity and user experience. During implementation, some complex interface components were generated with the assistance of Claude (Anthropic) in order to accelerate frontend development and dedicate more time to higher-level engineering tasks such as backend development, database design, API integration, model deployment, and system testing.

Finally, the different components of the application — frontend, backend, database, and machine learning models — were integrated into a unified platform. Extensive testing was conducted to verify the correctness of predictions, the reliability of the application, and the overall user experience before the platform reached its final deployment stage.

2.1 Global Architecture

The platform follows a layered client–server architecture. User interactions originate in the React frontend, which communicates with an Express.js REST API; the API delegates prediction requests to the trained machine learning models and persists results in a MongoDB database. The data flow can be summarized as follows:



This layered design cleanly separates presentation, business logic, and intelligence layers, allowing each layer to evolve independently — for instance, retraining a model does not require any change to the frontend.

2.2 Folder Structure

The project repository is organized into three top-level directories, each with a clearly defined responsibility:

- **frontend/** — contains all React components and pages. Isolating the frontend keeps the user interface decoupled from business logic, allowing independent development, styling, and testing of the client application.
- **backend/** — contains the data CSV files, trained machine learning models, Python scripts used to serve those models, database configuration, Mongoose schemas, controllers, routers, services, and utility functions. Grouping these together reflects the backend's role as the orchestration layer between data, intelligence, and the client.
- **notebooks/** — the starting point of the entire project, where everything was crafted, from exploratory data analysis to model training and evaluation. Keeping notebooks separate from production code preserves a transparent and reproducible research trail distinct from the deployed application.

2.3 Technologies

The project combines several technologies from the fields of data science, artificial intelligence, web development, and database management. Each technology was selected according to its strengths and its role within the overall architecture.

Technology	Purpose	Justification
Python	Machine learning and data analysis	Chosen for its simplicity and its rich ecosystem of AI and data science libraries.
Pandas	Data manipulation and preprocessing	Provides powerful tools for cleaning, transforming, and organizing datasets.
NumPy	Numerical computing	Enables efficient mathematical operations on large datasets.
Scikit-learn	Machine learning	Offers reliable implementations of preprocessing techniques, algorithms, and evaluation metrics.
SHAP	Model explainability	Explains the contribution of each feature to a prediction, improving model transparency.
Matplotlib	Data visualization	Used to generate statistical plots during the EDA phase.
Seaborn	Statistical visualization	Simplifies the creation of informative, aesthetically pleasing visualizations.

Technology	Purpose	Justification
Jupyter Notebook	Experimentation environment	Facilitates iterative data exploration, model development, and documentation in a single workspace.
CSV files	Dataset storage	Lightweight, portable, and directly compatible with most data science libraries.
React.js	Frontend development	Component-based architecture simplifies the creation of interactive, reusable user interfaces.
JavaScript	Frontend and backend programming	Allows the same language to be used across the entire web application.
HTML5	Page structure	Provides the structural foundation of the user interface.
CSS3	Styling	Responsible for the application's visual appearance and responsive design.
Node.js	Backend runtime	Executes server-side JavaScript efficiently using a non-blocking architecture.
Express.js	REST API	Simplifies backend development through lightweight routing and middleware management.
MongoDB	Database	A flexible NoSQL database well suited for storing user profiles and application data.
Mongoose	MongoDB object modeling	Simplifies database interactions through schemas and data validation.
REST API	Communication	Enables structured communication between the frontend and backend.
JSON	Data exchange	Lightweight format for transferring data between application layers.
Figma	UI/UX design	Used to prototype and design the application's interface before implementation.

Technology		Purpose	Justification
Visual Code	Studio	Development environment	Provides an efficient IDE with extensive extensions for web and Python development.
Git & GitHub		Version control & hosting	Tracks code modifications, facilitates project management, and supports portfolio presentation.

Table 2.1: Technologies used in the bacAidz platform and their justification.

2.4 Database Design

The application uses MongoDB as its primary database management system. A NoSQL, document-oriented database was preferred over a traditional relational database because most of the application’s data naturally fits a hierarchical JSON structure. This choice also integrates seamlessly with the MERN stack and provides the flexibility required to store prediction results, user information, and dynamically generated reports.

The database is composed of four main collections, each corresponding to a different stage of the application.

2.4.1 Users Collection

The Users collection stores the information required for user authentication and personalization. Each document contains the student’s identity, login credentials, academic stream (*filière*), and Baccalaureate session. Passwords are never stored in plain text; instead, they are automatically hashed using the BCrypt algorithm before being saved, ensuring secure authentication and protecting users against credential leakage. The collection also serves as the central entity of the database, as every prediction generated by the application is associated with a specific user.

2.4.2 Pre-BAC Results Collection

The Pre-BAC Results collection stores the outputs generated during the first phase of the platform. Each user owns a single Pre-BAC prediction containing the predicted Baccalaureate average, the cluster assigned by the clustering algorithm, the identified student archetype, the most similar student in the dataset, the corresponding similarity score, and the derived features computed during feature engineering. These results allow the application to preserve the student’s historical analysis while avoiding repeated model execution.

2.4.3 BAC Year Results Collection

The BAC Year Results collection is the largest and most detailed collection in the database. Instead of storing only the final average, each document contains a complete academic profile

covering every predicted subject. For each subject, the application stores the predicted grade, success probability, improvement potential, identified strengths, detected weaknesses, SHAP-based explanations, subject-specific derived features, and the original input values used during prediction. The collection also maintains global statistics summarizing the student's overall academic performance, including weighted average, strongest and weakest subjects, overall risk level, and cross-subject indicators generated by the machine learning pipeline.

2.4.4 Post-BAC Results Collection

The Post-BAC Results collection stores the personalized higher education recommendations generated by the orientation module. Each document contains the student's top ten recommended university programs, together with their ranking, predicted compatibility score, university information, domain, and a short description of each specialization. These recommendations are generated by combining machine learning predictions with a rule-based filtering system that considers admission thresholds and user preferences.

2.4.5 Relationships Between Collections

Although MongoDB is a non-relational database, logical relationships are maintained using document references. The Users collection acts as the central entity of the database: each prediction generated during the three phases references the corresponding user through a `user_id` field. This organization establishes a one-to-one relationship between a student and their Pre-BAC analysis, while allowing one user to accumulate multiple BAC Year predictions and Post-BAC recommendation sessions throughout the application's lifetime. Such a design ensures data consistency while preserving the flexibility offered by a document-oriented database.

Design Note — Embedding Subject Predictions

A deliberate design decision worth highlighting is that the fifteen Baccalaureate subjects are not represented as fifteen separate MongoDB collections. Instead, all subject predictions are embedded within a single `subjects[]` array inside the BAC Year Result document. This approach reduces the number of database queries, simplifies data retrieval, and naturally groups all subject predictions belonging to the same prediction session, balancing organization, performance, and maintainability rather than treating the schema as an afterthought.

2.5 UI/UX Philosophy

The design of the application goes beyond aesthetics. Every visual element, interaction, and navigation decision was made with one objective in mind: creating an interface that naturally resonates with Algerian Baccalaureate students. Since the target users are primarily teenagers between the ages of 16 and 19, the platform was designed to be modern, engaging, intuitive, and motivating throughout the entire academic journey.

2.5.1 Simplicity and Clarity

The interface follows a minimalist design philosophy where unnecessary visual clutter is avoided. Information is presented progressively, allowing students to focus on the most important tasks without feeling overwhelmed. Simple layouts, clear typography, and well-defined sections improve readability and make the platform accessible even to users with little technological experience.

2.5.2 Modern Color Palette

The application adopts a dark theme based primarily on shades of blue, violet, and white. This palette was selected for several reasons: a modern and professional appearance; reduced eye strain during long study sessions; strong visual contrast for improved readability; and a technological identity consistent with an AI-powered platform.

2.5.3 Arabic-First Experience

Although the application was developed using international technologies, the user interface primarily employs the Arabic language, since it is the language most familiar to Algerian Baccalaureate students and the language in which the official examinations are conducted. This choice reduces the learning curve and makes the application feel immediately familiar to its target audience.

2.5.4 Motivational Slogans

Short motivational slogans are integrated throughout the platform. Besides creating a more dynamic interface, these messages reflect an important aspect of Algerian student culture, where motivational phrases are widely shared during the Baccalaureate preparation period. Their objective is to maintain user engagement and create a more encouraging learning environment.

2.5.5 SVG Illustrations

The interface primarily relies on SVG illustrations rather than traditional photographs, offering lightweight files and faster loading times, infinite scalability without loss of quality, a consistent visual style, and easy customization to match the application's color palette.

2.5.6 Meaningful Icons and Terminology

Icons, labels, and technical terms were carefully selected to remain understandable for students unfamiliar with artificial intelligence or machine learning. Whenever possible, technical concepts are translated into simple educational language, allowing users to focus on the insights rather than the underlying algorithms.

2.5.7 Roadmaps and Progress Visualization

Several sections of the application present information as visual roadmaps, helping students better understand their current academic position, the objectives they should achieve, and the different stages of their Baccalaureate journey. Whenever chronological information is presented, such as academic progression or orientation procedures, the application uses visual timelines that improve comprehension by presenting information in a logical sequence.

2.5.8 Countdown Timer

The landing page includes a live countdown displaying the remaining days, hours, minutes, and seconds before the next Algerian Baccalaureate examination. Besides providing useful information, this element creates a constant sense of progression and reinforces motivation and planning.

2.5.9 Gamification

Several features of the platform extend beyond raw machine learning output to actively motivate students: prediction reports, strengths-and-weaknesses breakdowns, progress simulations, personalized recommendations, study plans, and automatically generated timetables. Rather than being purely informational, these features are designed to encourage continued engagement with the platform and measurable improvement over time, treating user motivation as a first-class design consideration rather than an afterthought.

2.5.10 User-Centered Design

Throughout the development process, every interface component was designed with one central question in mind: *“Would an Algerian Baccalaureate student immediately understand and enjoy using this feature?”* This philosophy guided not only the visual appearance of the application but also the organization of information, navigation flow, and interaction mechanisms, resulting in a platform that is both functional and engaging.

Chapter 3

The Pre-BAC Module

3.1 Objective

The Pre-BAC module was designed to investigate a series of recurring questions and widely shared beliefs surrounding the Algerian Baccalaureate. Every year, students, parents, teachers, and social media communities debate the factors that truly influence success in the BAC. While many opinions exist, very few are supported by objective data. The objective of this module was therefore to explore whether these commonly accepted ideas are merely part of the collective imagination or whether they can be observed through statistical analysis and machine learning. The main questions addressed by this module are summarized below.

1. Do previous academic results influence the Baccalaureate average?

One of the most frequently asked questions by Algerian students concerns the relationship between previous academic performance and BAC success. More specifically: does the Brevet average (the official national examination preceding high school) influence the final BAC average? Do grades obtained during the first and second years of secondary school (1AS and 2AS) correlate with BAC performance? Can students who performed poorly during these years still achieve a significant improvement — commonly referred to as a “*remontada*” — during the BAC? Conversely, do consistently high grades before the BAC guarantee success? This analysis aims to quantify these relationships rather than relying solely on anecdotal evidence.

2. Does the academic stream (*filière*) influence students’ grades?

Another common belief is that some academic streams systematically obtain higher or lower BAC averages than others. The objective was to determine whether the chosen *filière* represents a significant explanatory factor for academic performance and whether noticeable differences exist between streams.

3. Do students naturally form distinct academic profiles?

Beyond simple averages, this study investigates whether students within the same *filière* naturally group into clusters sharing similar academic characteristics. By applying clustering algo-

rhythms, the project seeks to identify common student profiles and determine whether meaningful patterns emerge from the data without predefined labels.

4. What is the influence of study habits and personal behavior?

Academic success is often associated not only with grades but also with behavioral factors such as study habits, time management, sleep, stress, consistency, and family support. This module explores the relationship between these behavioral variables and students' predicted academic performance to better understand their relative importance.

5. Can data help evaluate popular educational beliefs?

Finally, the project investigates several popular beliefs frequently discussed among Algerian students and on social media. One example concerns the widespread claim that students from the Kabylie region consistently achieve higher BAC success rates because of the so-called “*Saboura Sihria*” (“magic blackboard”), a popular expression referring to the alleged practice of teachers writing answers on classroom boards during examinations. Rather than accepting or rejecting such claims beforehand, the objective is to examine whether the collected data provides any evidence supporting these beliefs or whether they remain unsupported by the available observations.

Overall, the Pre-BAC module illustrates how data science can be used not only to make predictions but also to explore educational phenomena, challenge assumptions, and provide objective insights into the reality of the Algerian Baccalaureate system.

3.2 Dataset

3.2.1 Features

The Pre-BAC prediction model was built using a combination of demographic information, previous academic performance, study habits, behavioral characteristics, and engineered features, capturing as many factors as possible that may influence a student's final Baccalaureate performance. The features are grouped into the following categories.

3.2.2 Data Collection

Since no public dataset exists that accurately represents the academic journey of Algerian Baccalaureate students while containing all of the required variables, a custom dataset had to be constructed from several complementary sources.

Student surveys. Google Forms were designed and distributed among Algerian high school students, collecting demographic characteristics, previous academic performance, study habits, lifestyle and behavioral factors, perceived strengths and weaknesses, family support, and learning preferences. These responses provided the majority of the behavioral and academic variables used throughout the project.

Feature	Description
student_id	Unique identifier assigned to each student.
gender	Student’s gender.
region	Geographic region of residence (used to investigate the “Saboura Sihria” legend).
filiere	Academic stream followed by the student.
reason_for_choice	Main reason for choosing the academic stream (parents, academic guidance, grade pressure, friends, interest).
is_repeater	Indicates whether the student is repeating the BAC year.

Table 3.1: Student information features.

Brevet	First year (1AS)	Second year (2AS)
brevet_avg	as1_avg_global	as2_avg_global
brevet_math	as1_math	as2_math
brevet_physics	as1_physics	as2_physics
brevet_science	as1_science	as2_science
brevet_arabic	as1_arabic	as2_arabic
brevet_langues	as1_langues	as2_langues
brevet_hfada	as1_language_avg as1_hfada	as2_language_avg as2_hfada

Table 3.2: Pre-Baccalaureate academic performance features (Brevet, 1AS, 2AS).

Feature	Description
weekly_study_hours	Average study hours per week.
study_consistency	Regularity of study sessions (range 0 to 1).
time_essential_subjects	Time allocated to core BAC subjects.
time_non_essential_subjects	Time allocated to secondary subjects.
private_tutoring_hours	Weekly hours of private tutoring.
exam_anxiety	Self-reported examination anxiety level.
health_state	General health condition.
family_support_score	Perceived family support.
learning_style	Preferred learning strategy.
comprehension_speed	Speed of understanding new concepts.
focus_level	Ability to maintain concentration.

Table 3.3: Study habits and lifestyle features.

Official educational resources. To ensure that the generated datasets reflected the reality of the Algerian educational system, several official and widely recognized resources were consulted, including official Baccalaureate statistics, Ministry of Education publications, Ency-Education, DZ-Exams, official examination reports, and publicly available educational statistics. These resources were mainly used to understand examination structures, grading distributions, admission thresholds, subject coefficients, and historical educational trends.

Literature and domain knowledge. Some variables were selected based on findings commonly reported in educational research, particularly factors known to influence academic performance, such as study consistency, sleep habits, examination anxiety, and family support.

3.2.3 Data Generation

Although the collected data formed a solid foundation for the project, it was not sufficient on its own to train robust machine learning models: some variables were missing entirely, while others contained incomplete observations or were represented by only a limited number of samples. A controlled data generation process was therefore carried out to address these limitations. The objective was *not* to fabricate unrealistic information, but rather to generate additional observations that remained statistically coherent with the characteristics of Algerian Baccalaureate students observed during the collection phase.

Several constraints were respected during this process: generated grades remained within realistic educational ranges; relationships between previous and future academic performance were preserved; distributions remained consistent across the different *filières*; behavioral variables followed plausible combinations; and impossible or contradictory profiles were avoided. After generation, every dataset underwent an additional validation phase to ensure that the synthetic observations remained coherent with the original data before proceeding to model training. The final datasets contained approximately **1,500–2,000 student profiles**, providing sufficient diversity for the different machine learning tasks developed throughout the project.

3.2.4 Data Cleaning

Before performing any statistical analysis or training machine learning models, the collected and generated datasets underwent a comprehensive cleaning and preprocessing phase aimed at improving data quality, eliminating inconsistencies, and ensuring that the datasets accurately represented the academic profiles of Algerian Baccalaureate students. The main preprocessing operations are summarized below.

- **Missing values:** handled using strategies appropriate to each variable; records that could not be reliably estimated were removed to preserve the integrity of the dataset.
- **Duplicate records:** identified and removed to ensure that each observation represented a unique student.
- **Inconsistent data:** logically inconsistent values, such as impossible grade combinations or contradictory information, were corrected when possible or removed otherwise.

- **Outlier detection:** extreme observations were carefully analyzed to distinguish genuine exceptional performance from erroneous data entries; obvious errors were removed while legitimate high- or low-performing students were retained.
- **Data type verification:** numerical features were stored as numeric values, while categorical variables were encoded consistently.
- **Feature encoding:** categorical variables, including *filière*, learning style, gender, and region, were transformed into numerical representations suitable for machine learning algorithms.
- **Data validation:** a final pass verified that every feature respected its expected range and that relationships between variables remained coherent.

3.3 Exploratory Data Analysis

The dataset used in this study contains a representative sample of approximately 1,500 Algerian Baccalaureate students.

3.3.1 Distribution Across Streams and Success Rates

The Sciences Expérimentales stream is by far the most represented in the dataset. This popularity can be explained by its balanced curriculum, which combines scientific subjects with relatively well-distributed coefficients while offering students a wide range of higher education opportunities after the Baccalaureate. In contrast, the other streams generally lead to a more specialized set of university programs and career paths, making them less attractive to a large proportion of students.

The Mathematics stream achieves the highest success rate, exceeding 80%. A possible explanation is that this stream is highly selective, attracting students with particularly strong mathematical and scientific backgrounds; although it contains fewer students than the Sciences Expérimentales stream, its population tends to exhibit a higher overall academic performance, supporting the common perception that the Mathematics stream prioritizes academic selectivity over quantity.

3.3.2 Grade Distributions

The distribution of Brevet, 1AS, 2AS, and Baccalaureate average grades across the different *filières* reveals that the distribution of Baccalaureate averages is approximately normal, centered around a score of 10/20, while still containing a number of high- and low-performing outliers. In contrast, the distributions of the Brevet, 1AS, and 2AS averages exhibit a slight left skew, indicating a higher concentration of students with relatively good grades during the years preceding the Baccalaureate.

Comparing these distributions suggests that the Baccalaureate examination introduces greater variability in student performance, indicating that a considerable number of students

experience significant changes in their academic results during the BAC year, whether positive or negative. In particular, this provides evidence that strong improvements (“remontadas”) are indeed possible: previous academic performance should not be considered the sole determinant of success, offering an encouraging perspective for students aiming to improve their final results.

3.3.3 Grades by Stream and Region

A combination of violin plots and boxplots was used to study the distribution, central tendency, variability, and exceptional observations of Baccalaureate grades according to both academic stream and geographic region (wilaya).

Performance across academic streams. The Sciences Expérimentales and Techniques Mathématiques streams exhibit the largest variability, reflecting a more heterogeneous student population with both high- and low-performing students. The Mathematics stream displays a different pattern: most students are concentrated within a relatively narrow range around the passing threshold, resulting in a more compact distribution, while also containing numerous high-performing outliers approaching 20/20, consistent with the selective nature of this stream. The remaining streams — Gestion et Économie, Langues Étrangères, and Lettres et Philosophie — show relatively symmetric distributions centered around the passing threshold, with median grades close to 10/20.

Performance across regions. Unlike the differences observed between academic streams, the distributions across the different wilayas are remarkably similar: the median Baccalaureate average lies between 10 and 11 out of 20 for almost every region, following a comparable bell-shaped pattern, with the majority of students obtaining grades between approximately 9 and 12.5. Every region also contains a number of high-achieving students whose averages exceed 16/20, demonstrating that excellent academic performance is not concentrated within a particular geographic area.

Discussion. Overall, these results suggest that the academic stream has a much greater influence on the distribution of Baccalaureate grades than the student’s geographic region. While students from different wilayas exhibit broadly similar performance patterns, each *filière* possesses its own distinctive distribution, reflecting differences in student selection, curriculum structure, and academic requirements.

3.3.4 Correlation Analysis

The correlation heatmap provides an overview of the linear relationships between the different features and the final Baccalaureate average. The Second-Year Secondary School (2AS) average exhibits the strongest positive correlation with the Baccalaureate average, reaching a correlation coefficient of approximately **0.86**, indicating that academic performance during the year immediately preceding the BAC is the best predictor of the final examination result. The 1AS average and the Brevet average also show significant positive correlations, both close to **0.67**, though noticeably weaker than the 2AS average, suggesting that more recent academic performance carries greater predictive power.

In contrast, most behavioral variables exhibit relatively weak linear correlations with the Baccalaureate average. Variables related to study habits, stress, family support, and lifestyle generally present low correlation coefficients when considered individually. This should not be interpreted as evidence that these factors are unimportant: their effects are often indirect or involve complex nonlinear interactions that cannot be fully captured by simple pairwise correlation analysis, and are better modeled by machine learning algorithms capable of learning nonlinear patterns, such as the Random Forest Regressor used in this project.

3.3.5 Previous Performance vs. Baccalaureate Results

A scatter plot of 2AS average against final Baccalaureate average, colored by *filière*, reveals a remarkably strong positive linear relationship: students who achieved high averages during their 2AS year generally maintained similar levels of performance in the Baccalaureate, while students with lower previous averages tended to obtain lower BAC results. This relationship remains consistent across all academic streams. A closer inspection also reveals differences between streams: Sciences Expérimentales and Mathematics occupy a wider range of the performance spectrum and include the majority of the highest-achieving students, while Lettres et Philosophie students are more concentrated around intermediate grades.

3.3.6 Study Hours and Academic Performance

A scatter plot of weekly study hours against final Baccalaureate average, colored by study consistency, reveals a counter-intuitive finding: studying more hours does not necessarily result in better academic performance. The highest Baccalaureate averages are generally observed among students studying approximately **8 to 17 hours per week**, provided that they maintain a **high level of study consistency**. Conversely, students reporting significantly longer study durations, often exceeding 20 hours per week, tend to obtain lower average grades and generally exhibit lower consistency scores.

One possible interpretation is that the quality and regularity of studying are more influential than the total number of hours invested: students who study consistently throughout the academic year may require fewer hours to achieve excellent results because their learning process is more organized and efficient, whereas very high reported study hours may reflect inefficient study strategies or periods of intensive last-minute revision rather than sustained learning. Overall, these findings suggest that academic consistency and previous academic performance are considerably stronger indicators of Baccalaureate success than the simple quantity of weekly study hours alone.

3.3.7 The “Saboura Sihria” Hypothesis: Kabylie vs. Non-Kabylie

This analysis investigates a widespread belief among Algerian students that the Kabylie region consistently achieves higher academic performance in the Baccalaureate examination. A clear difference can be observed between the two distributions: the median Baccalaureate average for students from the Kabylie region is noticeably higher, reaching approximately **11.25/20**,

whereas the median for the Non-Kabylie group is slightly above the passing threshold, around **10.30/20**. For the Kabylie group, nearly the entire interquartile range lies above the official passing threshold of 10/20, suggesting that the majority of students achieved passing grades, whereas the lower quartile of the Non-Kabylie distribution falls below the passing threshold. The lower whisker is also positioned higher for the Kabylie group, meaning that even the lowest-performing students in that group generally achieved better results than their Non-Kabylie counterparts. Both groups, however, contain numerous high-performing outliers exceeding 16/20, demonstrating that exceptional academic performance is not exclusive to a particular region.

Overall, the analysis suggests that students from the Kabylie region exhibit a higher overall level of academic performance within the collected dataset. This observation should nonetheless be interpreted with caution: the present analysis identifies a statistical difference between the two groups but does not establish its underlying causes. Factors such as socioeconomic conditions, educational resources, school quality, or cultural influences were not directly measured and therefore cannot be inferred from this analysis alone.

3.3.8 Validation Against Official 2024 Results

The success rates observed in the constructed dataset were compared against the official success rates published for the 2024 Algerian Baccalaureate examination — the year in which the author completed the BAC. The overall distribution of success rates across the different *filières* closely matches the official statistics. Although slight differences are expected due to sampling variability, the general trends remain highly consistent, and the national Baccalaureate success rate generally remains around 60%, making the generated dataset representative of the overall Algerian educational context. This comparison provides an additional validation of the dataset, suggesting that it reasonably reflects the real distribution of student outcomes observed at the national level.

3.4 K-Means Clustering Analysis

To better understand the different academic profiles inside each *filière*, a K-Means clustering algorithm was applied separately to the six streams, using the features most strongly correlated with the Baccalaureate average. The optimal number of clusters for each stream was selected using both the Elbow Method and the Silhouette Score, balancing cluster compactness and separation. After clustering, each resulting group was assigned a meaningful Arabic archetype name, later used to improve the user experience of the matching system described in Section 3.6.

3.4.1 Discussion

Several observations can be drawn from this clustering analysis. Sciences Expérimentales exhibits the greatest diversity of student profiles, ranging from very low performers to outstanding students, and is the only stream where the majority of students (nearly 59%) belong to clusters whose average exceeds the passing threshold. The Mathematics stream remains one of

Filière	Students	Clusters	Key Observation
Sciences Expérimentales	630	6	Richest diversity of profiles, ranging from a Very Low cluster averaging $\sim 8/20$ to an Excellent cluster averaging more than $17/20$; $\sim 41\%$ in the two lowest clusters, $\sim 37\%$ in Good/Very Good/Excellent.
Mathematics	150	4	More than 56% of students fall in the Failing cluster despite the stream containing some of the highest national averages, confirming its demanding and selective nature.
Techniques Mathématiques	165	3	$\sim 1/3$ in the Average cluster, more than half in the Failing group, only a limited share reaching Good.
Gestion et Économie	225	4	Lower performance ceiling (highest cluster average $\sim 13.6/20$); nearly three quarters of students in the Passable and Failing clusters.
Langues Étrangères	180	3	More than half in the Failing cluster, $\sim 1/3$ in Average, only a small share reaching Good.
Lettres et Philosophie	150	3	$\sim 53\%$ Failing, $\sim 37\%$ Average, $\sim 10\%$ Good — a relatively homogeneous, average-centered distribution.

Table 3.4: K-Means clustering results per academic stream.

the most selective, producing exceptional students while simultaneously presenting a high proportion of struggling learners (only 12.7% of students belong to the two highest-performing clusters, while more than 56% fall into the lowest-performing cluster). Techniques Mathématiques follows a similar trend, with roughly half of its students in satisfactory clusters and 53.3% in the weakest cluster. Gestion et Économie shows a more evenly distributed population, with approximately 35% of students below the 10/20 threshold. Finally, Langues Étrangères and Lettres et Philosophie display almost identical distributions, with more than half of students in the lowest-performing cluster.

Overall, the clustering confirms that students do not form a single homogeneous population within each stream. Instead, each *filière* contains distinct academic profiles that can be automatically identified through unsupervised learning. These clusters later serve as the basis of the student matching system implemented in the Pre-BAC module, allowing each user to be compared with students sharing similar academic characteristics.

3.5 Rule-Based Logic: The Student Profiler

To transform raw predictions into a complete, personalized profile, a rule-based `analyze_student()` function was implemented. Given a student identifier, it returns a comprehensive, Arabic-language profile combining academic performance, behavioral patterns, risk assessment, and actionable recommendations.

3.5.1 Basic Information and Clustering Context

The function first extracts the student’s *filière*, cluster identifier (0–4 depending on the stream), academic archetype (e.g., “*Balanced Achiever*”, “*Struggling Grinder*”), percentile rank within the *filière*, and a comparison against the cluster average. Clustering groups students with similar academic profiles, making comparisons more meaningful than comparing a student to the entire *filière* average.

3.5.2 Derived Features

The function computes more than 13 derived features that reveal hidden patterns in the data, summarized in Table 3.5.

Each derived feature is automatically interpreted with a clear Arabic label, the calculated value, a human-readable message explaining its meaning, and a status badge (excellent, good, or needs work).

3.5.3 Subject-by-Subject Analysis

The function compares the student’s grade in each subject against their cluster average, classifying the result as a strength, a level consistent with the cluster, or a weakness, with the magnitude of the difference reported explicitly. This contextualizes raw grades rather than presenting them in isolation, making it immediately clear where a student is excelling or struggling relative to comparable peers.

3.5.4 Risk Assessment

A composite risk score (0–100) is computed from multiple weighted factors, as summarized in Table 3.6. Three risk levels are then reported: Low Risk (0–29, student is on track), Medium Risk (30–59, some concerning factors), and High Risk (60–100, multiple issues requiring intervention).

3.5.5 Personalized Recommendations and What-If Scenarios

Based on the computed profile, the system generates actionable recommendations across several categories: BAC level improvement, study efficiency enhancement, sleep optimization, weakness targeting, stress management, and burnout (“grinder”) prevention. Each recommendation includes a concrete action, supporting details, and an estimated expected improvement

Feature	Formula	Interpretation
Study Efficiency	$\text{bac_avg} / (\text{study_hours} + 0.1)$	Points earned per hour of study (>1.2 = excellent)
Grinder Index	$\text{study_hours} / (\text{efficiency} + 0.1)$	Hours needed to gain one point (>30 = burnout risk)
Improvement Momentum	$\text{as2_avg} - \text{as1_avg}$	Grade trajectory (positive = improving)
Grade Volatility	$\text{std}(\text{subject_scores})$	Consistency across subjects (<1.0 = stable)
Sleep Debt	$\max(0, 8 - \text{sleep_hours})$	Hours of sleep deficit (>1.5 = problematic)
Health–Stress Index	$(6 - \text{health_state}) \times \text{stress_level}$	Combined health and stress burden
Weakness Count	$\text{count}(\text{weak_subjects} > 0.6)$	Number of subjects with significant weakness
Problem Solving Gap	$(\text{weak_math} + \text{weak_physics}) / 2$	Weakness in analytical subjects
Science vs. Language Gap	$\text{science_weak} - \text{lang_weak}$	Disparity between scientific and language skills
Subject Balance	$\text{std}([\text{math}, \text{physics}, \text{science}, \text{arabic}, \text{languages}])$	Overall subject consistency
Consistent Weak	$\text{consistency} \times \text{weakness_count}$	Hard work without results (ineffective effort)
Efficient Stressed	$\text{efficiency} \times \text{stress_performance}$	Burnout risk indicator

Table 3.5: Derived features computed by the Pre-BAC student profiler.

Risk factor	Weight	Condition
Low BAC score	20–40	$<10 \rightarrow 40, <12 \rightarrow 20$
Multiple weaknesses	15–25	$>3 \text{ weaknesses} \rightarrow 25, >2 \rightarrow 15$
Sleep deprivation	15	Sleep debt >2 hours
High exam anxiety	15	$\text{stress_performance} > 0.45$
Negative momentum	20	$\text{improvement_momentum} < -0.5$
Ineffective effort	15	$\text{efficiency} < 0.6 \text{ AND } \text{hours} > 20$

Table 3.6: Composite risk score factors.

in points. The function additionally simulates what-if scenarios, estimating the potential impact of specific interventions: fixing weak subjects (estimated +0.6 to +2.0 points), improving study efficiency through active recall and spaced repetition (+1.2 points), or improving sleep

to a consistent 8 hours for two weeks (+0.8 points).

3.5.6 Output and Practical Value

The function returns a single structured profile combining basic information, risk assessment, derived insights, subject-by-subject analysis, strengths and weaknesses, recommendations, what-if scenarios, and underlying statistics. By combining machine learning (clustering, feature engineering) with educational-psychology concepts (study efficiency, burnout risk, sleep impact), the Student Profiler transforms raw student data into actionable intelligence that is useful not only to students — to understand their strengths and weaknesses — but potentially to teachers and guidance counselors seeking to identify at-risk students and provide targeted support.

3.6 Student Matching and Frontend Experience

The frontend implementation of the Pre-BAC module is a multi-step interactive form that collects comprehensive student data across six progressive stages, then processes this data through a matching engine and derived-feature computation to generate personalized insights.

3.6.1 Input Collection

The system collects data through six structured steps: (1) basic information — academic stream, region, repeater status, and reason for choosing the stream; (2) Brevet grades in Mathematics, Physics, Science, Arabic, and Foreign Languages; (3) First-Year Secondary (1AS) grades for the same five subjects; (4) Second-Year Secondary (2AS) grades, the strongest predictor of BAC results; (5) study habits and lifestyle — weekly study hours, study consistency (0.1–1.0), sleep hours (4–9), stress level (1–5), exam anxiety, family support, and private tutoring hours; and (6) a final review and submission step.

3.6.2 Cluster Assignment

The student’s cluster is determined using a rule-based approach driven by their predicted BAC score, with stream-specific thresholds: Sciences Expérimentales uses six clusters (e.g., $\geq 17 \rightarrow$ Cluster 0 through $< 9 \rightarrow$ Cluster 5), Mathematics uses three clusters (≥ 17.5 , $14.5\text{--}17.5$, < 14.5), and the remaining streams use three clusters (≥ 14 , $11.5\text{--}14$, < 11.5). Each cluster has a pre-calculated average BAC score and a detailed Arabic description explaining the student’s profile, strengths, weaknesses, and specific recommendations.

3.6.3 Matching Engine

The system finds the closest matching student from the real dataset using a weighted similarity algorithm, summarized in Table 3.7. Each feature difference is normalized to a 0–1 scale and weighted to compute an overall similarity score (lower is more similar); the top five matches are displayed, and the closest match is selected.

Feature category	Weight	Features
2AS grades	40%	Math, Physics, Science, Arabic, Languages
1AS grades	15%	Math, Physics, Science
Brevet grades	10%	Math, Physics, Science
Study habits	20%	Hours, consistency, sleep
Lifestyle	15%	Stress, anxiety, family support

Table 3.7: Weighted similarity components of the student matching engine.

3.6.4 Hybrid Prediction Method

The final BAC score displayed to the student is calculated using a hybrid approach combining the matching engine and a regression formula:

$$\text{Final Score} = (\text{Match Score} \times 0.5) + (\text{Regression Score} \times 0.5)$$

where the Match Score is the actual BAC score of the closest matching student, and the Regression Score is a weighted formula based on the student’s own grades and habits: the 2AS average (55%), the 1AS average (20%), the Brevet average (10%), the Math score (5%), the Science score (5%), and the Languages score (5%), with additional adjustments for sleep, study efficiency, consistency, stress, family support, tutoring, region, and repeater status.

3.6.5 Derived Features and Visualization

The frontend computes 20 derived features extending the backend’s analysis (study efficiency, grinder index, time allocation ratio, resource reliance, improvement momentum, grade volatility, stress-performance ratio, sleep debt, health-stress index, weakness count and concentration, science-language gap, subject balance, *filière* alignment, repeater penalty, family support, and a regional “Kabylie boost” indicator), and generates two visualizations: a radar chart comparing the student’s grades in each subject against their cluster average, and a study habits comparison chart against ideal targets.

3.6.6 User Experience

The interface features a progress bar showing completion status (1/6 to 6/6), step-by-step navigation with back/next buttons, real-time Arabic descriptions and explanations, color-coded feedback (green for good, red for areas needing improvement), detailed cluster explanations in accessible Arabic, and actionable recommendations with their expected improvement in points.

Chapter 4

The BAC Year Module

This chapter presents the core component of the project: the BAC Year module. Its objective is to assist students throughout their Baccalaureate preparation by combining machine learning predictions with explainable AI techniques and an interactive user interface.

4.1 Overview

The application is organized around clickable subject cards, where each card redirects the user to a dedicated page for the selected subject. Every subject page contains several components designed to provide both predictions and practical guidance. First, the student is introduced to the official examination structure, including exam duration and format according to the selected *filière*, familiarizing students with the nature of the examination before moving to the prediction stage. Next, the student fills in a form containing the most influential features identified during exploratory data analysis for that particular subject; these features are passed to a dedicated machine learning model that predicts the student's expected score.

To make the prediction interpretable rather than a simple numerical output, SHAP (SHapley Additive exPlanations) is used to identify the student's strengths and weaknesses, analyzing the contribution of every feature to the predicted score and translating it into understandable feedback. The application also includes a simulation engine built upon the correlations discovered during exploratory analysis: by modifying specific input variables, students can simulate potential improvements and immediately observe their estimated impact on the predicted subject average. Finally, a comprehensive report summarizes all generated information, including the predicted score, identified strengths and weaknesses, simulated improvements, and personalized recommendations.

Since the Algerian Baccalaureate contains fifteen different subjects, a separate machine learning pipeline was developed for each: fifteen datasets, fifteen Jupyter notebooks, and fifteen trained models. Although each subject has its own specific features and dataset, all notebooks follow the same methodological pipeline: data preparation, exploratory data analysis, feature engineering, model selection, training, evaluation, explainability, and deployment. To avoid making this report unnecessarily long and repetitive, the common methodology is presented only once in this chapter; the specific characteristics, features, and results of each subject are summarized in dedicated tables.

Beyond subject prediction, this module also integrates several rule-based systems: a resource recommender suggests learning materials according to the student’s predicted performance and detected weaknesses, a to-do list generator creates personalized study tasks based on the same information (fully customizable by the user, since user comfort remains a priority), and the generated tasks are transformed into a personalized study timetable using an optimization-based scheduling algorithm. Each of these components is described in detail later in this chapter.

4.2 The Per-Subject Machine Learning Pipeline

Although fifteen independent pipelines were developed, all of them follow an identical methodological structure, summarized in Table 4.1.

Stage	Description
Imports	Standard data science stack: Pandas and NumPy for manipulation; Matplotlib and Seaborn for visualization; Scikit-learn (<code>RandomForestRegressor</code> , <code>MultiOutputRegressor</code> , <code>train_test_split</code> , evaluation metrics) for modeling; SHAP for explainability.
Dataset generation	Collecting real data for ~2,000 students across all 15 BAC subjects would have required an impractical amount of time and resources. Datasets were instead generated from raw educational materials collected from official Algerian resources — syllabuses, examination structures, national statistics, and historical educational documents — which served as the foundation for realistic, subject-specific datasets.
Export / Load	The generated dataset is exported as a CSV file for persistence, then reloaded into a Pandas DataFrame for analysis and modeling.
EDA overview	Dataset shape (~500 rows, ~22 columns), data types, descriptive statistics, missing value and duplicate checks, and memory usage.
Score distribution & normality	Histogram of the BAC score distribution with mean/median lines and passing rate; Q–Q plot, boxplot, and Shapiro–Wilk / Kolmogorov–Smirnov normality tests; skewness and kurtosis analysis.

Stage	Description
Correlation heatmaps	Used to identify the input features with the strongest predictive power for the target subject grade, and to provide the statistical foundation for the improvement simulation engine, which relies on these correlations to generate realistic, data-driven improvement scenarios.
Feature distributions	Histograms for trimester grades, subject-specific skills, and generic study habits, each with summary statistics.
Feature engineering	Creation of derived features from the original skill scores, specific to each subject, later explained to the student through a dedicated frontend section.
Train/test split	80%/20% split with <code>random_state=42</code> for reproducibility; the trimester average is retained as a feature, being the strongest individual predictor; train/test index overlap is explicitly verified.
Model training	A <code>MultiOutputRegressor</code> wrapping a <code>RandomForestRegressor</code> (100 trees, <code>max_depth=10</code>) predicting three outputs simultaneously: the BAC score (0–20), the success probability (binary, $\geq 10/20$), and the improvement potential (0–10 points).
Detailed evaluation	See Section 4.3.
SHAP explainability	See Section 4.4.
Weakness analysis	Features with negative SHAP values are identified as weaknesses, features with positive SHAP values as strengths; SHAP values are also aggregated across the dataset to identify the most common weaknesses per subject.
Simulation engine	Real impact factors derived from the correlation analysis are used to estimate the effect of input changes on the predicted score.

Stage	Description
Student-friendly explanations	All technical outputs are translated into accessible Arabic explanations, with performance indicators (Strong / Average / Needs Improvement) and plain-language descriptions of derived features such as language proficiency, analytical ability, practice intensity, and subject balance.

Table 4.1: Common methodological pipeline applied to all fifteen BAC Year subject models.

4.3 Model Evaluation

The performance of each subject-specific model was assessed using several complementary evaluation metrics to ensure both predictive accuracy and practical reliability. Since fifteen independent models were trained, the values reported below correspond to the **average performance observed across all subjects**.

Metric	Average Value
Mean Absolute Error (MAE)	0.74
Mean Squared Error (MSE)	0.89
Root Mean Squared Error (RMSE)	0.94
Coefficient of determination (R^2)	0.895

Table 4.2: Average regression performance across the fifteen subject-specific models.

These results indicate that the models are capable of accurately estimating students' expected Baccalaureate grades, explaining nearly 90% of the variance in the data. To evaluate the practical usefulness of the predictions, the regression outputs were also transformed into a pass/fail classification using the official passing threshold of 10/20. The resulting confusion matrices report true positives, true negatives, false positives, and false negatives, complemented by a full classification report (precision, recall, F1-score) evaluating the balance between correctly identifying successful and unsuccessful students.

Several visualizations were produced to facilitate model interpretation and validation: an Actual vs. Predicted scatter plot assessing the agreement between predicted and real grades; a residual distribution histogram verifying that prediction errors remain randomly distributed around zero; and a feature importance chart highlighting the variables that contribute most to each model's predictions.

Finally, an error analysis was performed to evaluate the practical accuracy of the predictions: on average, **78%** of all predictions differed from the actual grade by less than one point, while **92%** had an error of less than two points. These results demonstrate that the proposed models provide sufficiently accurate predictions for educational guidance and personalized decision support.

4.4 Explainability with SHAP

Machine learning models often provide highly accurate predictions but may behave as “black boxes,” making it difficult to understand the reasoning behind a specific prediction. To overcome this limitation and increase the transparency of the platform, SHAP (SHapley Additive exPlanations) was integrated into the application. SHAP assigns an importance value to every input feature by measuring its contribution to the final prediction, allowing the system to explain *why* a student obtained a given prediction rather than only displaying a numerical score.

For each trained subject model, three complementary SHAP visualizations are generated:

- **Feature importance bar plot:** the average contribution of each feature across the entire dataset, identifying the variables with the greatest influence on subject performance.
- **SHAP beeswarm summary plot:** the distribution of SHAP values for every feature across all students, simultaneously highlighting feature importance, the direction of influence (positive or negative), and the variability of impact across different student profiles.
- **Waterfall plot:** a detailed explanation for an individual student’s prediction, showing how each feature contributes to moving the predicted grade upward or downward from the model’s baseline prediction until the final estimated score is obtained.

These SHAP-derived explanations are exploited throughout the application to detect students’ strengths and weaknesses, generate personalized reports, recommend appropriate learning resources, and support the simulation engine, transforming the predictive model from a simple forecasting tool into an interactive decision-support system capable of providing meaningful, interpretable feedback.

4.5 Interactive Notebook Experience

Before being deployed in the web application, every subject pipeline was prototyped as an interactive experience inside its Jupyter notebook using `ipywidgets`, allowing the prediction logic to be validated end-to-end before frontend integration. This interactive layer included: a dashboard introducing the BAC exam structure for the subject (duration, total points, sections per stream, and color-coded skill importance); a multi-section input form (stream selection, trimester grades, subject-specific skills, study habits, and psychological factors) using sliders, radio buttons, and dropdown menus; a prediction output view presenting the predicted score, performance level (Excellent, Very Good, Good, Acceptable, or Needs Improvement), success probability, improvement potential, and a skill breakdown with progress bars; a weakness analysis view comparing the student’s profile against target levels across skills and study habits, producing an overall weakness score (0–100); a plain-language explanation of all derived features; an interactive simulation panel with eleven sliders (six skills and five habits) reflecting real correlations, allowing students to see the real-time impact of adjustments on their predicted

score; and a final, professional report consolidating the score summary, student information, key metrics, strengths and weaknesses, a detailed skills breakdown, a personalized action plan with five or more recommendations, and a closing summary with overall advice.

4.6 Subjects and Features

Table 4.3 summarizes the original and derived features used for each of the fifteen Baccalaureate subjects modeled by the platform. While the specific skills naturally differ by subject (e.g., grammar and rhetoric for Arabic versus mechanics and electricity for Physics), every subject consistently includes trimester grades, subject-specific competencies, study-habit indicators, and psychological factors such as confidence and stress.

Subject	Representative Original Features	Representative Derived Features
Arabic	trimester grades, grammar, essay, poetry, prose, rhetoric, comprehension, practice intensity, study habits, confidence, stress	language_core_score, writing_score, analysis_score, imbalance_score, practice_intensity
French	comprehension textuelle, inference, thesis/argument identification, essay structure, language accuracy, historical/argumentative text analysis	comprehension_score, argumentation_score, writing_score, text_type_score, practice_intensity, global_skills_score
English	main idea detection, inference, skimming/scanning, grammar transformation, vocabulary, essay structure, coherence/cohesion	reading_score, language_score, writing_score, grammar_mastery, overall_proficiency, study_efficiency
Langue Étrangère	reading comprehension, grammar transformation, translation skill, paragraph structure, idea development	reading_score, linguistic_score, writing_score, practice_intensity, overall_proficiency
Tamazight	narrative understanding, morphology, sentence decomposition, narrative/descriptive writing, cultural heritage	language_core_score, writing_score, analysis_score, imbalance_score, practice_intensity

Subject	Representative Original Features	Representative Derived Features
Mathematics	functions, algebra, probability, sequences, complex numbers, geometry, integral calculus, proof reasoning, exam time management	analysis_block, algebra_block, probability_block, complex_block, geometry_block, imbalance_score, psychological_composite
Physics	mechanics, electricity, chemistry (general/esterification/acid-base), nuclear, waves/oscillations, problem solving	mechanics_block, electricity_block, chemistry_block, nuclear_block, waves_block, imbalance_score
Science (SVT)	molecular biology, immunology, energy (photosynthesis/respiration), neuroscience, geology, document analysis	molecular_block, protein_block, immune_block, energy_block, neuro_block, scientific_skills_composite
Techno	mechanics/RDM, gear transmission, automation/GRAFCET, logic circuits, structural analysis, technical drawing	mechanics_block, automation_block, chemistry_block, civil_block, overall_skill_score
His_Geo	historical/dates/figures memory, geography knowledge, document analysis, essay method, Cold War / decolonization / Algeria history	memory_composite, methodology_composite, knowledge_composite, hg_imbalance
Philo	dialectical/argumentative/comparative method, conceptual understanding, logical reasoning, critical thinking	methodology_composite, thinking_composite, writing_composite, philo_imbalance
Islamia	Quran recitation, tafsir, hadith comprehension/analysis, fiqh (ibadah/muamalat), aqida, document analysis	quran_composite, hadith_composite, fiqh_composite, methodology_composite, overall_islamia

Subject	Representative Original Features	Representative Derived Features
Economie	market mechanisms, money/banking, unemployment, inflation, graph interpretation, situation analysis	economics_composite, management_composite, practice_intensity, market_focus
Gestion	calculation accuracy, financial logic, cost/financial/loan blocks, execution speed	gestion_imbalance, practice_intensity, study_quality, overall_level
Loi/Droit	company law, labor law (individual/collective), public finance, legal reasoning, case exercises	legal_skills_composite, labor_law_composite, overall_law_score, reasoning_focus

Table 4.3: Subjects and representative features used across the fifteen BAC Year models.

4.7 Cross-Subject Observations

Beyond the individual model evaluations, several recurring patterns emerged from the exploratory analysis conducted across all fifteen subjects.

- Grade distributions across most subjects closely follow a normal distribution, indicating a realistic representation of student performance.
- Some subjects within the Gestion-Économie and Lettres-Philosophie streams exhibit a slight left-skewed distribution, suggesting that a larger proportion of students achieve relatively higher grades in these subjects.
- Subjects such as Mathematics and Physics show stronger correlations with behavioral and psychological factors — study habits, consistency, and stress level — highlighting the importance of non-academic features in predicting performance.
- The trained models perform very well for the majority of students, accurately predicting the scores of average profiles; prediction errors tend to increase for extreme outlier cases, which is expected given the limited representation of such profiles in the training data.
- Some subject grade distributions, particularly in the Gestion-Économie stream, exhibit a ceiling effect where high grades are more frequent and concentrated, naturally reducing the variability of the target variable and making it harder to distinguish between top-performing students.

- For almost all subjects, the strongest correlated feature with the target grade was the average of the three trimester grades, confirming that recent, sustained academic performance is the single most informative predictor available, consistent with the Pre-BAC correlation analysis.

4.8 Rule-Based Resource Recommender

The Resource Recommender and the To-Do List Generator described in this and the following section do not use machine learning models; instead, they apply explicit, pre-defined rules grounded in educational research and teaching experience, operating directly on the outputs of the trained prediction models.

The Resource Recommender uses a priority-based filtering system to suggest the most relevant educational resources — YouTube channels, books, Facebook groups, and apps — for each student. Resources are first filtered by stream, so that, for example, Mathematics students never see Economics resources. Each subject is then assigned a priority level based on its predicted score and the number of detected weaknesses, following the rule:

$$\text{Priority} = \begin{cases} \text{Critical} & \text{if score} < 10 \text{ or weaknesses} \geq 2 \\ \text{High} & \text{if score} < 12 \text{ or weaknesses} \geq 1 \\ \text{Medium} & \text{if score} < 14 \\ \text{Low} & \text{otherwise} \end{cases}$$

Resources are displayed in order of priority, Critical first, ensuring students are guided toward the subjects that need the most urgent attention before being shown supplementary material for subjects they have already mastered.

4.9 Rule-Based To-Do List Generator

The To-Do List Generator creates a personalized weekly study plan based on the student's academic profile, stream, and predicted performance. Each subject is assigned a coefficient-based score (1–5, derived from its official coefficient within the student's stream) and an ML-based adjustment derived from the predicted score, combined as:

$$\text{Priority} = \text{round}((\text{Coefficient Score} \times 0.9) + (\text{ML Adjustment} \times 0.1))$$

where the ML adjustment ranges from +1.5 for a predicted score below 10 down to −1.5 for a predicted score of 15 or above, ensuring that both the official importance of a subject and the student's actual need for improvement are reflected in the resulting priority.

Each subject is then assigned one or two tasks (two if priority ≥ 4), randomly selected among four task types — Review, Exercises, Past BAC papers, and Weakness Focus — each carrying a base duration (1.0, 1.5, 3.0, and 1.5 hours respectively) further adjusted by coefficient and ML-score multipliers, then rounded to the nearest half hour and clamped between 0.5 and 4.5

hours per task. If the resulting weekly workload exceeds the safety cap of 45 hours, non-priority tasks are progressively reduced before Past BAC tasks are shortened, ensuring that no task ever drops below 0.5 hours. The generated list remains fully customizable by the student, who can edit, remove, or reorder tasks, since respecting individual study preferences was treated as a core design priority rather than an afterthought.

4.10 Timetable Generation

4.10.1 Design Goal

Once a weekly to-do list has been generated, the platform must place each task into a concrete weekly schedule: a 7-day $\times$ 7-time-slot grid (8:00 to 22:00, Saturday to Thursday, matching the Algerian school week), respecting hard constraints such as no overlapping tasks, a maximum of three tasks per day, a minimum one-hour rest period between sessions, and a 45-hour weekly cap, while optimizing for soft objectives such as scheduling tasks at their preferred time of day, spreading tasks evenly across the week, and prioritizing high-importance tasks in higher-quality time slots.

4.10.2 Implementation: A Greedy, Genetic-Algorithm-Inspired Heuristic

The deployed implementation uses a deterministic, priority-driven greedy placement strategy rather than a full evolutionary search: tasks are sorted by priority and assigned, in order, to the best currently available slot according to a weighted scoring function that mirrors the components of a genetic algorithm’s fitness function — preferred-time alignment, daily spread, duration fit, and rest balance — after which a constraint-repair pass resolves any residual overlap, daily-load, or rest-period violations by shifting tasks to the next valid slot. This design was directly inspired by a project completed during the Introduction to Artificial Intelligence course at ENSIA, in which search techniques such as Constraint Satisfaction Problems (CSP), Hill Climbing, and Simulated Annealing were applied to generate valid academic timetables.

Engineering Note

A full population-based Genetic Algorithm — chromosome encoding of the weekly grid, tournament selection, two-point crossover, swap mutation, and elitism over multiple generations — was designed conceptually and is documented below as the natural next iteration of this module, but the version currently deployed in production is the lighter-weight greedy heuristic described above. This distinction is stated explicitly here for transparency, since the underlying optimization concepts (fitness components, hard/soft constraints, repair mechanism) are shared between both versions and the genuine GA upgrade is listed as concrete future work in Section 8.

4.10.3 Designed Genetic Algorithm (Future Iteration)

The conceptual GA design that the greedy heuristic emulates, and that constitutes the natural next implementation step, is specified as follows.

Chromosome representation. Each chromosome encodes a complete weekly timetable as a 7×7 grid (7 days, 7 two-hour time slots between 8:00 and 22:00), where each gene is either a task assignment (subject, duration, priority, preferred time) or an empty slot.

Population. An initial population of 100 individuals is generated through random valid timetable construction, ensuring all tasks from the to-do list are placed while respecting the no-overlap and maximum-three-tasks-per-day constraints.

Fitness function. Each candidate timetable is scored using a weighted combination of five components: priority placement (30%, rewarding the assignment of high-priority tasks to high-quality slots), time alignment (25%, rewarding tasks scheduled at their preferred time of day), spread (20%, rewarding an even distribution of tasks across the week), duration fit (15%, rewarding tasks that fit cleanly within their assigned slot), and rest balance (10%, rewarding adequate gaps between consecutive study sessions).

Genetic operators. Parent selection uses tournament selection with a tournament size of three; reproduction uses two-point crossover with an 80% probability, exchanging a contiguous day/time segment between two parent timetables; mutation uses swap mutation with a 5% probability per gene, randomly exchanging the position of two tasks; and elitism preserves the top 10% of each generation unchanged to guarantee monotonic improvement of the best solution found.

Constraint handling. Hard constraints (no overlap, maximum three tasks per day, task duration within slot bounds, minimum one-hour rest period, 45-hour weekly cap) are enforced through a repair mechanism applied after crossover and mutation, while soft constraints (preferred time, even distribution, priority scheduling, subject variety) are optimized through the fitness function itself rather than being strictly enforced.

Termination. The algorithm would terminate after a maximum of 500 generations, when the fitness plateaus for 100 consecutive generations without improvement, or when a near-optimal fitness threshold of 0.95 is reached, whichever occurs first.

Aspect	Greedy (de- ployed)	Genetic Algorithm (de- signed)
Solution quality	Good, may be suboptimal	Near-optimal
Speed	Fast, $O(N^2)$	Slower, $O(G \times P \times N \times T)$
Global search	No	Yes
Constraint handling	Manual repair	Built-in via repair operator
Adaptability	Fixed rule ordering	Evolves over generations

Table 4.4: Comparison between the deployed greedy heuristic and the designed Genetic Algorithm.

Chapter 5

The Post-BAC Module

5.1 Motivation

Many Algerian students spend thirteen years of schooling chasing grades, in part to prove their worth to their families, to the point of losing sight of the final purpose of those years: choosing a higher-education orientation that genuinely suits them. Others struggle simply because they do not yet know their vocation or have no particular field of interest. The Post-BAC module exists to address this gap, helping students navigate the orientation process with concrete, personalized information rather than vague advice.

5.1.1 The Algerian Higher Education System

The Algerian higher education system is centrally managed by the Ministry of Higher Education and Scientific Research (MESRS) and organized into eleven major academic domains: Sciences and Technologies (ST); Mathematics and Computer Science (MI); Medicine and Paramedical Sciences; Sciences of Nature and Life (SNV); Economic, Management and Commercial Sciences (SEGC); Law and Political Sciences (DSP); Human and Social Sciences (SHS); Arabic Language and Literature (LLA); Foreign Languages (LLE); Islamic Sciences; and Arts, Culture and Sports (STAPS).

Students access these domains through several distinct educational pathways. The most common is the **LMD system** (Licence–Master–Doctorat), adopted by the majority of Algerian universities, following a standardized, credit-based structure from Bachelor’s through Master’s and eventually doctoral studies. For students pursuing engineering careers, admission is generally offered through **National Higher Schools** (Écoles Supérieures), requiring a two-year preparatory cycle followed by a competitive entrance examination, after which successful candidates continue with a three-year engineering program leading to a state-recognized diploma equivalent to a Master’s degree. The **Écoles Normales Supérieures (ENS)** specialize in training future teachers for the national education system. Finally, professional health programs — Medicine, Pharmacy, Dentistry, and Paramedical studies — follow a **Long Cycle** rather than the LMD system, typically lasting six to seven years and focusing on intensive professional training. This diversified organization allows students to pursue academic or professional careers through pathways adapted to their interests while remaining within a unified,

state-regulated system.

5.2 Knowledge Base Structure

To support this module, a comprehensive JSON knowledge base was built, organizing all available higher-education formations in Algeria as an array of domain objects. Each domain object contains the domain name, a brief description, the list of eligible Baccalaureate streams, and a domain-specific weighted-average formula (for example, the Mathematics and Computer Science domain uses $(\text{bac_avg} \times 2 + \text{maths_grade})/3$).

Within each domain, formations are organized into up to four categories: **ENS** programs (teacher training, including recruitment type, interview requirement, and accepted streams with priority levels), **Écoles Supérieures** (elite specialized schools, including location, duration, accepted streams, specializations, and career outcomes), **Ingénieur** (university-based engineering programs), and **LMD** (Bachelor programs). For every formation, the **streams_acceptes** field maps each eligible Baccalaureate stream to an admission priority level (P1 highest, P2 secondary, P3 lower) and a minimum required weighted average, based on 2025 admission data. This structure enables stream-based filtering, domain-specific weighted-average calculation, eligibility determination against minimum thresholds, priority-based recommendation, and the display of comprehensive program information (duration, location, specializations, and career outcomes) for all eleven domains.

5.3 Dataset

Six datasets were collected, one per Baccalaureate stream, since orientation outcomes depend strongly on the stream chosen and the courses taken during that stream; each dataset contains roughly 500 students. The features used include the official Baccalaureate average and stream, the grades obtained in all fifteen BAC subjects, the three trimester averages, and behavioral indicators such as stress level, competitiveness, consistency score, career orientation, and wilaya. The target variable is **domaine_cible**, one of the eleven academic domains available in Algeria. From these raw grades, the domain-specific weighted average for each of the eleven domains was subsequently computed using the formulas described in the previous section.

5.4 Exploratory Data Analysis

The exploratory analysis for this module examined the distribution of grades for each stream and subject, confirming that students generally perform better in their stream's essential subjects, and that the sample was constructed so that all students had passed the Baccalaureate (a prerequisite for choosing a higher-education orientation). It also re-confirmed, in this new context, the variability of performance among different streams already discussed in the Pre-BAC chapter. Six correlation heatmaps, one per stream, were produced to study the relationship between domain-specific weighted averages and target orientations, illustrating each stream's relative effectiveness across the eleven domains and the average admission threshold of each

domain. The repartition of all students by their Baccalaureate mention, both overall and per stream, was also analyzed, along with the repartition of streams across target domains, presented through both bar charts and heatmap visualizations.

5.5 Machine Learning Model and Evaluation

The post-BAC orientation system employs a specialized ensemble of **six Random Forest Classifiers**, one trained per Baccalaureate stream (Sciences, Maths, Technique Mathématique, Gestion, Langues, Lettres), to predict the most suitable academic domain for each student. Each model is configured with 200 decision trees, a maximum depth of 12, and a minimum of 15 samples per split, balancing predictive accuracy and generalization. The models are trained on a feature set of 18 engineered variables, including domain-specific weighted subject scores (e.g., `weighted_maths_info`, `weighted_st`, `weighted_medecine`), academic ratios (`ratio_maths_science`, `ratio_maths_physique`), stream alignment metrics (`scientific_score`, `literary_score`), and behavioral indicators (`consistency_score_final`, `avg_variability`).

Each stream-specific model is trained on 500 student profiles (80% training, 20% testing) and achieves accuracy scores ranging from **85% to 95%**, demonstrating the effectiveness of the stream-specific modeling approach over a single global classifier. In production, when a student submits their Baccalaureate results, the system identifies their stream, selects the corresponding model, computes the 18 features, and predicts the three most appropriate academic domains for that student's profile.

5.6 Rule-Based Orientation Engine

The machine learning predictions are combined with a sophisticated rule-based engine that transforms a student's predicted domains into concrete, actionable program recommendations through a multi-stage filtering and prioritization process. For each student, the engine computes domain-specific weighted averages using the tailored formulas described above (for example, the Medicine domain uses a weighted formula incorporating science, mathematics, and physics with different coefficients), then applies a cascade of filters: stream eligibility; comparison of the student's weighted average against each domain's minimum admission threshold; orientation filtering for ENS programs based on the student's stated career goal (teaching versus research); and location-based filtering for LMD programs, restricting results to formations available in the student's own wilaya. Finally, all eligible formations across the predicted domains are globally ranked by their minimum admission threshold in descending order, ensuring that the most selective and prestigious programs are surfaced first.

This hybrid design — machine learning to identify promising domains, combined with transparent, explainable rules to check eligibility — ensures that students receive recommendations that are not only academically relevant to their profile, but also practically feasible given their actual grades, geographic location, and career aspirations.

5.7 Frontend Integration

From a technical perspective, the Post-BAC module is served, like the rest of the platform, through the MERN stack, with predictions and the resulting top-ten ranked program list stored in MongoDB and associated with the requesting user. Rather than presenting a flat list of program names, the system explains the reasoning behind each suggestion — the domain match, the computed weighted average, and the specific admission threshold satisfied — allowing students to understand why a particular orientation suits their academic profile rather than simply being told what to choose.

Chapter 6

Backend Architecture

6.1 Overview

The backend of the bacAidz platform is a RESTful API built with Node.js and Express.js, designed to handle all business logic, data persistence, and machine learning integration. It follows a Model-View-Controller (MVC) architectural pattern, ensuring a clean separation of concerns, maintainability, and scalability.

6.2 API Architecture

6.2.1 RESTful Design and Base Structure

The API follows REST principles, using standard HTTP methods and status codes to handle client requests. All endpoints are organized around resources (students, subjects, results) and support CRUD-style operations through predictable URL patterns: **GET** requests retrieve data (user results, subject lists), while **POST** requests create or process data (predicting a BAC score, submitting student data). The API is organized under a common base URL, grouped into logical resource namespaces: `/api/auth/` for authentication, `/api/user/` for user management, `/api/bacyear/` for BAC prediction and analysis, `/api/subjects/` for subject metadata, and `/api/results/` for student results.

6.2.2 Dynamic Routing

The system uses dynamic route parameters (`:subject`) to handle all fifteen subjects through a single, reusable route definition rather than fifteen near-duplicate routes:

```
1 // One route handles all subjects
2 router.post('/:subject/predict', predict);
3 // Subjects: arabic, maths, physics, science, french, english,
4 // his_geo, islamia, philo, techno, tamazight,
5 // langue_etrangere, droit, economie, gestion
```

Routes are organized into dedicated files by resource: `authRoutes.js` (login, register, logout), `userRoutes.js` (profile management), `bacYearRoutes.js` (prediction, analysis, results), and `subjectRoutes.js` (subject metadata and resources).

Endpoint	Method	Description
/api/auth/register	POST	User registration
/api/auth/login	POST	User authentication
/api/bacyear/:subject/predict	POST	Predict BAC score for a subject
/api/bacyear/:subject/weaknesses	POST	Analyze student weaknesses
/api/bacyear/:subject/features	POST	Get derived features
/api/bacyear/:subject/simulate	POST	Simulate improvement scenarios
/api/bacyear/results	GET	Get all user results
/api/bacyear/:subject/result	GET	Get specific subject result

Table 6.1: Key API endpoints exposed by the bacAidz backend.

6.3 Controllers

Controllers form the core business logic layer: they validate incoming requests, orchestrate calls to the service layer, persist results, and return formatted JSON responses. A representative controller for a prediction request is shown below.

```

1 // backend/src/controllers/bacYearController.js
2 const predict = async (req, res) => {
3   try {
4     const { subject } = req.params;
5     const userData = req.body;
6     const userId = req.user.id;
7
8     // Validate required fields
9     const requiredFields = ['grade_t1', 'grade_t2', 'grade_t3'];
10    for (const field of requiredFields) {
11      if (userData[field] === undefined || userData[field] === null) {
12        return res.status(400).json({
13          success: false,
14          message: 'Field ${field} is required'
15        });
16      }
17    }
18
19    // Load model and run prediction
20    await bacYearService.loadModel(subject);
21    const prediction = await bacYearService.predict(subject, userData);
22    const derivedFeatures = bacYearService.calculateDerivedFeatures(subject,
23      userData);
24    const weaknessAnalysis = bacYearService.analyzeWeaknesses(userData,
25      subject);
26
27    // Persist and respond
28    await saveResult(userId, subject, userData, prediction, derivedFeatures,

```

```

    weaknessAnalysis);
27   res.json({ success: true, data: { prediction, derivedFeatures,
    weaknessAnalysis, subject } });
28
29   } catch (error) {
30     res.status(500).json({ success: false, message: 'Prediction error',
    error: error.message });
31   }
32 };

```

Controller responsibilities are deliberately kept consistent across the codebase: input validation, authentication delegation to middleware, subject-alias mapping (e.g., `loi` → `droit`), orchestration of the appropriate services in the correct order, database operations, response formatting, and centralized error handling. Four controllers cover the entire application surface: `authController.js` (register, login, logout, token refresh), `userController.js` (profile retrieval/update, password change), `bacYearController.js` (prediction, weaknesses, features, simulation, results), and `subjectController.js` (subject list, metadata, resource management).

6.4 Middleware and Authentication

Middleware functions execute before route handlers to perform authentication, logging, validation, and error handling. The authentication middleware extracts a JSON Web Token (JWT) from the `Authorization` header, verifies it, and attaches the corresponding user document to the request object before allowing the request to proceed:

```

1  // backend/src/middleware/auth.js
2  const protect = async (req, res, next) => {
3    try {
4      let token;
5      if (req.headers.authorization?.startsWith('Bearer ')) {
6        token = req.headers.authorization.split(' ')[1];
7      }
8      if (!token) return res.status(401).json({ success: false, message: '
    Please log in first' });
9
10     const decoded = jwt.verify(token, process.env.JWT_SECRET);
11     const user = await User.findById(decoded.id).select('-password');
12     if (!user) return res.status(401).json({ success: false, message: 'User
    not found' });
13
14     req.user = user;
15     next();
16   } catch (error) {
17     return res.status(401).json({ success: false, message: 'Invalid session,
    please log in again' });
18   }
19 };

```

The authentication flow proceeds as follows: a user registers, their password is hashed with `bcrypt` and stored; on login, credentials are verified and a JWT is issued; the client stores this token and attaches it to subsequent requests; protected routes verify the token via the `protect` middleware before granting access. JWT payloads contain the user identifier, username, email, *filière*, issue time, and a 7-day expiry. Passwords are hashed with `bcrypt` using 12 salt rounds, and rate limiting is applied globally to mitigate brute-force attacks. Authentication-related routes (`/api/auth/login`, `/api/auth/register`) are public, while all application routes (`/api/bacyear/*`, `/api/user/profile`) require a valid JWT.

6.5 Project File Structure

The backend follows a layered directory structure reflecting the MVC pattern and its supporting concerns:

```

1 backend/
2 | -- src/
3 |   | -- config/           # Database connection, environment variables
4 |   | -- controllers/      # Request handlers (auth, user, bacYear)
5 |   | -- middleware/       # Auth, error handling, validation
6 |   | -- models/           # Mongoose schemas (User, BacYearResult, ...)
7 |   | -- routes/           # Express route definitions
8 |   | -- services/         # Business logic (prediction, analysis)
9 |   | -- utils/            # Helpers and validators
10 |   | -- ml_models/        # Serialized scikit-learn models (.pkl, x15)
11 |   | -- ml_scripts/       # Python inference scripts (predict.py)
12 |   |-- app.js             # Express application entry point
13 |-- package.json
14 |-- .env
15 |-- server.js             # Server startup, database connection

```

6.6 Technology Stack Summary

6.7 End-to-End Data Flow: A Prediction Request

To illustrate how the layers of the backend interact, consider a single Mathematics prediction request, traced end-to-end:

1. The frontend issues `POST /api/bacyear/maths/predict` with the student's grades and skill inputs in the request body.
2. The authentication middleware verifies the JWT and attaches the authenticated user to the request.
3. The controller validates the required fields (`grade_t1`, `grade_t2`, `grade_t3`) and delegates to the service layer.

Layer	Technology	Purpose
Runtime	Node.js	JavaScript runtime environment
Framework	Express.js	Web framework for building REST APIs
Database	MongoDB	NoSQL database for storing application data
ODM	Mongoose	MongoDB object modeling and schema validation
Authentication	JWT + bcrypt	Stateless authentication and password hashing
Machine Learning	Python + scikit-learn	Model training and inference
Integration	Child process	Python–Node.js communication for inference
Security	Helmet, CORS	Security headers, cross-origin resource sharing

Table 6.2: Backend technology stack summary.

4. The service layer loads the serialized Mathematics model from `ml_models/bac_year/maths_model.pkl` and invokes the Python inference script through a child process.
5. The Python script loads the model, preprocesses the input, runs the `RandomForestRegressor` prediction, and returns a JSON result.
6. The service layer maps the Python output to the frontend’s expected format and computes derived features and weakness analysis.
7. The result is persisted to the `BacYearResult` collection in MongoDB, and the user’s summary statistics are updated.
8. The controller returns a structured JSON response containing the prediction, derived features, and weakness analysis.

6.8 Conclusion

The backend architecture follows established software engineering practices, with a clean separation of concerns that makes the system maintainable, scalable, and secure. The integration of Node.js with Python for machine learning combines the strengths of both ecosystems: a robust, asynchronous web framework for API management, and a mature scientific computing stack for predictive analytics. The JWT-based authentication, MVC pattern, and service-oriented design together produce a backend capable of supporting the full range of features described throughout this report.

Chapter 7

Limitations

Although the proposed platform achieved promising results, several limitations should be acknowledged honestly.

7.1 Synthetic Data Generation

Due to the absence of publicly available datasets covering all the required features, a significant portion of the data was generated based on official educational resources, statistical reports, and observed academic trends. Although extensive exploratory analysis was performed to ensure realistic distributions, generated data cannot perfectly reproduce the complexity of real student behavior.

7.2 Dataset Size

The datasets contain between 1,500 and 2,000 samples, which is sufficient for this project's objectives but remains relatively small compared to industrial-scale machine learning applications. Increasing the number and diversity of samples would likely improve the robustness and generalization of the models.

7.3 Possible Bias

The collected and generated data may contain different forms of bias related to region, stream distribution, socioeconomic background, or feature generation assumptions. Consequently, predictions should be interpreted as informed estimations rather than absolute truths.

7.4 Random Forest Limitations

Random Forest models provide excellent predictive performance and robustness; however, they do not extrapolate well outside the range of the training data, are less interpretable than simple linear models, and may produce less accurate predictions for rare or exceptional student profiles.

7.5 Prediction of Exceptional Students

The trained models perform very well for average student profiles but are generally less accurate for extreme cases, such as exceptionally high-performing or struggling students, since these profiles are naturally underrepresented in the training datasets.

7.6 Evolution of Admission Thresholds

Admission thresholds for Algerian universities and higher schools change every academic year depending on the number of candidates, available seats, and decisions made by the Ministry of Higher Education. Consequently, the orientation recommendations generated by the platform should be considered indicative rather than official.

7.7 Limited Feature Set

Although a large number of academic and behavioral features were considered, several factors that may influence student performance — such as family income, school quality, teacher effectiveness, and unexpected personal circumstances — were intentionally omitted because reliable data could not be collected.

7.8 Rule-Based Components

Some components of the platform, including the resource recommender, to-do list generator, and orientation explanations, rely on manually designed rule-based systems. While these rules were built using educational knowledge and observed patterns, they may not perfectly reflect every student's individual needs.

7.9 Timetable Optimization

As discussed in Chapter 4, the timetable generator currently deployed in production uses a greedy, priority-driven heuristic rather than a full population-based Genetic Algorithm. While the heuristic produces valid, constraint-respecting schedules, it does not perform a global search over the solution space and may converge to a locally, rather than globally, optimal weekly timetable.

Chapter 8

Future Work

Although the proposed platform already provides a comprehensive set of features, several improvements can be explored to further increase its accuracy, usability, and scalability.

8.1 Collection of Real-World Data

A primary future objective is to replace the synthetic datasets with large-scale datasets collected from real Algerian Baccalaureate students over multiple academic years, improving the reliability and generalization capability of the predictive models.

8.2 Genuine Genetic Algorithm Implementation

As discussed in Chapter 4, the conceptual Genetic Algorithm design for the timetable generator — chromosome encoding, tournament selection, two-point crossover, swap mutation, and elitism — is fully specified but not yet implemented; replacing the current greedy heuristic with this genuine evolutionary search is a concrete and well-scoped next step.

8.3 Deep Learning Models

Future versions of the platform could investigate deep learning approaches, such as neural networks, to model more complex relationships between academic, behavioral, and psychological features and student performance.

8.4 Integration of Large Language Models

Large Language Models could be integrated to provide intelligent academic assistance, answer students' questions, explain predictions in natural language, generate personalized study plans, and offer interactive guidance throughout the Baccalaureate preparation process.

8.5 Automatic Synchronization with Official Data

The platform could be connected to official Ministry of Higher Education resources or APIs, when available, to automatically update admission thresholds, university programs, and orientation information each academic year.

8.6 Student Accounts and Progress Tracking

Future versions could expand student accounts to monitor progress over time, compare different simulations, and receive recommendations informed by the student's full learning history rather than a single prediction session.

8.7 Cloud Deployment

Deploying the application on cloud platforms would improve accessibility, scalability, and performance while allowing simultaneous use by thousands of students during the Baccalaureate period.

8.8 Mobile Application

A dedicated Android and iOS application would make the platform more accessible to students, allowing them to access predictions, recommendations, and study tools directly from their smartphones.

8.9 Real-Time Analytics Dashboard

Future work may include a real-time analytics dashboard capable of aggregating anonymous student data to identify national trends, monitor performance across regions and streams, and provide valuable insights for students, teachers, and educational institutions.

8.10 Recommendation System Enhancement

The resource recommender could be expanded using hybrid recommendation techniques that combine machine learning with collaborative filtering, providing highly personalized educational resources based on students with similar profiles.

8.11 Continuous Model Improvement

As new data becomes available, the predictive models could be periodically retrained and optimized using more advanced ensemble methods, automated hyperparameter tuning, and continuous performance monitoring to maintain high prediction accuracy.

Conclusion

This report presented the design and development of bacAidz, a complete machine learning platform dedicated to the Algerian Baccalaureate. Its objective was not only to predict students' academic performance, but to transform complex data science techniques into an intuitive and accessible tool capable of assisting students throughout their entire Baccalaureate journey.

The platform was structured around three complementary phases. **Pre-BAC** analyzes students' previous academic history and identifies hidden patterns within the Algerian educational system, using statistical analysis and K-Means clustering to investigate and, in several cases, confirm or nuance widely held beliefs about the Baccalaureate. **BAC Year**, the core of the platform, predicts subject-specific performance across all fifteen Baccalaureate subjects, identifies strengths and weaknesses through SHAP-based explainability, recommends learning resources, generates personalized study plans, and produces optimized study timetables. **Post-BAC** assists students in selecting the higher-education domains and university programs that best match their academic profile and career aspirations, combining stream-specific Random Forest classifiers with a transparent rule-based eligibility engine.

Throughout this work, every stage of a typical data science pipeline was addressed: data collection and generation, preprocessing, exploratory data analysis, feature engineering, model training and evaluation, explainability through SHAP, rule-based reasoning, backend development, database design, and the implementation of a modern, culturally adapted web interface using the MERN stack.

Although the project presents several limitations, particularly regarding the use of synthetic data, the dynamic nature of university admission thresholds, and the gap between the deployed greedy timetable heuristic and its designed Genetic Algorithm successor, the obtained results demonstrate that machine learning can effectively capture meaningful patterns within the Algerian Baccalaureate ecosystem and provide valuable decision-support tools for students.

Beyond its technical contribution, this project represents a personal learning journey. It allowed me to combine the mathematical foundations, programming skills, web development knowledge, and machine learning concepts acquired throughout my studies at ENSIA into a single real-world application, built independently over eight weeks. More importantly, it showed me, concretely, how artificial intelligence can be applied to address an educational challenge that I lived through myself, and transform raw data into a practical, user-friendly solution for the students who will face that same challenge next.